
A Judge That Cannot Certify Itself: A Code-Enforced Trust Protocol for Adversarial LLM Auditing

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Abstract

1 LLM-based evaluators fail most dangerously when they are *reliable without being*
2 *valid*: internally consistent, confident, and biased. The largest systematic study of
3 LLM-as-a-judge to date reports exactly this dissociation—test–retest consistency
4 above 0.95 coexisting with severe position bias, and exact-match agreement over-
5 stating discrimination by 33–41 points of κ once chance-corrected [1]. Such a
6 judge is at its most harmful the moment it is allowed to *certify itself*—to declare
7 its own output validated, or to count its own internal agreement as independence.
8 We present a *trust protocol*: a small set of invariants an adversarial LLM-audit
9 pipeline is made to obey *in ordinary code*, not in prompts, and each pinned by a
10 test. (1) The pipeline can never return VALIDATED on internal grounds; human
11 closure is a *cryptographic* HMAC of the audit-ledger digest under a key the model
12 cannot reach, so a model that authors the payload cannot sign its own validation.
13 (2) Cross-model independence is *attested by the calling adapter*, not read from
14 a self-report. (3) A run is admitted only under an *A+B contract*—a *measured*
15 minimum of layers actually executed **and** every other layer carries an explicit
16 applicable/not-applicable verdict—otherwise the pipeline *refuses to close* and ex-
17 its non-zero, a refusal that overrides even a valid human attestation. (4) The engine
18 is turned on itself and the audit trail is shipped, including a different-vendor re-
19 view that overturned an earlier version of our own central empirical claim. We
20 make **no detection-superiority claim**; the contribution is the closure discipline
21 itself, and a per-run *disclosure record*—which layers ran, which were adjudicated
22 out and why, what independence was attested, whether closure was reached—that
23 a judge cannot silently violate. This is a machine-checked instance of the judge-
24 deployment disclosure standard JUDGE is convening [4].

25 1 The problem: self-certification, not inaccuracy

26 Evaluation validity is a property of a judge *in a system*, not in isolation [4]. A well-calibrated
27 judge can fail systematically once its output gates a safety decision or feeds back into training. The
28 specific failure we target is *self-certification*: a judge that treats its own stability as validity and
29 its own agreement as independence. Norman et al. [1] supply the empirical warrant—across 21
30 judges from nine providers and roughly 541,000 individual judgments, reliability and validity are
31 dissociated—but a diagnosis is not a mechanism. If a biased-but-consistent judge can stamp its own
32 output “validated,” everything downstream inherits an error the system cannot see. The question we
33 take up is not *how accurate is the judge* but *can the judge be stopped from closing its own loop*.

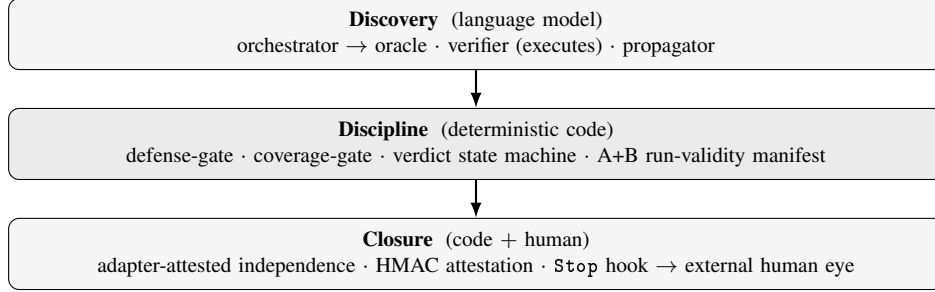


Figure 1: Three bands. Fallibility lives in the language-model band; *discipline and trust live in ordinary code*. The core assigns verdicts and decides closure; it can refuse, but it cannot self-certify.

2 The protocol

The system is an orchestrator that spawns adversarial LLM sub-agent roles (an oracle that cites facts and emits no verdicts; a verifier that recomputes and, where code is runnable, *executes* it; a propagator that constructs the concrete sequence violating a guarantee) and hands their findings to a *deterministic core* that assigns verdicts and decides completion. The orchestrator coordinates; it does not judge (Figure 1). Four invariants live in that core.

I1 — Non-closure is cryptographic. The completion state machine reaches `VALIDATED` only on a valid HMAC of the ledger digest under an operator key held outside the model’s reach. A bare “human-approved” token that does not verify closes nothing. A `Stop` hook re-verifies the HMAC in the operator’s environment and, if a `VALIDATED` does not verify, downgrades the ledger on disk. The model authors the payload but not the key: it cannot forge its own validation.

I2 — Independence is attested, not claimed. Different-vendor review is credited (independence level 3) only from the identity the *calling adapter* reports. A different-vendor label present only in the payload is recorded as `CROSS_MODEL_CLAIMED` and buys no independence credit. This directly answers the self-preference / correlated-blind-spot facet: a second model prompted adversarially by the same vendor is not counted as an independent draw.

I3 — The A+B run-validity contract. A pipeline that silently skips layers can still print a confident verdict; the danger is a run that *looks* complete because no one counted what did not happen. A run is admitted only if **(A)** a measured minimum of required layers actually executed—measured from each layer’s own output, not from self-report—and **(B)** every other layer carries an explicit `RAN/NOT_APPLICABLE/MISSING` status with justification. The manifest computes `run_validity`; if it is not `VALID`, completion is forced to `INVALID_RUN`, which *overrides every other state, including a valid human VALIDATED*, and the process exits non-zero. Closure cannot be bought by signing an incomplete process. The required minimum was itself *measured* across ten runs over eight artifact classes, not assumed—and the measurement corrected an earlier over-inclusion (one role frozen as required was returned to optional once two classes reached their defects without it).

I4 — Self-audit, shipped. The engine is run against its own artifacts and papers and the ledgers are committed. This is not decoration: a same-vendor self-audit (independence level 1) located four defects in our own closure guarantees; a later different-vendor review (level 3) overturned the original headline empirical claim and forced two guarantees from convention into cryptographic code; and the measurement instrument of I3, when first run, found three real defects *in itself*. We report these because a system whose value is adversarial honesty must show the audits that caught it.

Each invariant is a test in a 173-test suite run in CI across Python 3.10–3.13, including an end-to-end step that *fails if the pipeline ever prints VALIDATED without a human*.

Reproducibility and artifact. The system is an MIT-licensed engine with four reproducible benchmarks and the 173-test suite above. From a clone, `python3 -m unittest discover -s`

```

run_validity : VALID # A+B contract satisfied
completion  : EXTERNAL_REVIEW_PENDING # no attested external identity → not
validated
layers      : triage=RAN oracle=RAN verifier=RAN propagator=RAN governor=RAN
              reasoner=NOT_APPLICABLE # not selected by triage
              deep_causal=NOT_APPLICABLE # not selected by triage
              external_auditor=NOT_APPLICABLE # not selected by triage
source_grade : {primary:2, institutional:0, generalist:0, unknown:0}

```

Figure 2: A per-run *disclosure record* — verbatim output of `run_core.py` on a complete run. It states which layers ran, which were adjudicated out and *why*, that the A+B contract holds (VALID), and that closure was *not* reached because no external identity was attested. This is a machine-checked instance of the disclosure JUDGE is convening; the pipeline refuses to close when the record is incomplete.

72 tests and reproduce.py --strict on each benchmark recompute every quantitative claim to
73 the digit; the disclosure record of Figure 2 is the verbatim output of `run_core.py`. An anonymized
74 copy of the repository is provided for review so that reviewers can clone it and run the gates, the
75 refusal, and the cryptographic-closure check themselves; the public repository is disclosed on ac-
76 ceptance.

77 3 Relation to existing work

78 **Reliability vs. validity of judges.** [1] is the motivation and the missing-enforcement counterpart:
79 they measure that judges are reliable-without-valid and recommend a “Minimum Viable Validation
80 Protocol,” but a recommendation is advisory—a judge can decline it and certify itself anyway. We
81 make the most dangerous consequence, self-certification, structurally impossible in code. We do not
82 re-measure judge bias.

83 **Adversarial defect discovery.** The closest adversarial system, Refute-or-Promote [2], shares the
84 adversarial kill-mandate and a cross-model critic and demonstrates them at scale (a 31-day cam-
85 paign, ~171 candidates, ~79% killed before disclosure, real CVEs and accepted ISO C++ defect
86 reports). We claim no priority on those ideas. The difference is the objective: Refute-or-Promote
87 optimizes *precision of discovery* with a human orchestrator and no enforced non-closure—its cross-
88 model step *improves* reliability but nothing *forbids* a confident close. We optimize *trustworthiness*
89 *of closure*: judging is code, independence must be adapter-attested, closure is cryptographic, and an
90 under-run is refused non-bypassably. A better *finder* versus a *finder that cannot lie about having*
91 *finished*; the two are complementary.

92 **Adversarial robustness of judges.** Work on prompt-injection attacks against judges [3] asks
93 whether a judge can be *fooled from outside*. We ask the orthogonal question—whether a judge
94 can be *stopped from certifying itself from inside*—and enforce the answer in code rather than in a
95 prompt.

96 **JUDGE’s disclosure template.** The workshop’s community deliverable is a judge-deployment
97 disclosure standard (provenance, deployment context, known failure modes, human validation) [4].
98 Our execution manifest and A+B record are a *machine-checked instance* of such a disclosure: emit-
99 ted per run, refused when incomplete.

100 4 Limitations

101 We make no detection-superiority claim; a small-sample study behind this system (seven real errors,
102 one run per model nature) is a descriptive dissociation, not a recall result, and is reported as such
103 in the companion systems paper. The cryptographic non-closure is only as strong as the operator’s
104 key hygiene—it contracts forgery from a payload field to an out-of-band secret, it does not eliminate
105 it. Real cross-vendor independence depends on an adapter the operator must wire up; absent it, the
106 external-auditor role is theatre, and the code says so. And the whole apparatus is a *survivor*: every

107 internal audit sits at independence level ≤ 3 in a single lab and validates nothing—by construction,
108 only an external human eye closes the loop. That is the protocol’s own thesis applied to itself.

109 **References**

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- 115 [3] N. Maloyan and D. Namiot. Adversarial Attacks on LLM-as-a-Judge Systems: Insights from
116 Prompt Injections. arXiv:2504.18333, 2025.
- 117 [4] JUDGE @ NeurIPS 2026. Can We Trust the Judge? Workshop call for papers and failure
118 taxonomy. <https://judge2026.github.io/>, 2026.

119 Anonymized for double-blind review. Artifact (MIT-licensed engine; four reproducible benchmarks; 173 tests; CI-checked
120 protocol invariants) available at an anonymized repository; de-anonymized on acceptance.